

An exact algebraic envelope for four-by-four Ritz sets with an interior eigenvalue

Candidate partial result for arXiv:2004.05288

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Status. Candidate substantial partial result, likely valid, pending human review. The theorem completely treats the first missing 4×4 case under the stated nondegeneracy assumptions; the source's broad questions for arbitrary dimension and for $k \geq 3$ remain open.

Abstract

Dela Rosa and Woerdeman ask for a characterization of $\text{conv } \mathcal{B}_A(\mu)$ when a normal matrix has eigenvalues in the interior of its numerical range. We derive an explicit real-rooted support polynomial in all dimensions. When A has four distinct noncollinear eigenvalues and $\mu \notin \sigma(A)$, its weight polytope is a segment and the polynomial has bidegree at most $(2, 2)$. In every direction the exact support is attained at an endpoint root, an interior simple stationary root, or an interior multiple root. These candidates are algebraic; generically the two interior searches reduce to quartics. Intersecting the resulting supporting half-planes gives $\text{conv } \mathcal{B}_A(\mu)$ exactly. This covers spectra with an interior eigenvalue, where the polygonal formula of the source can fail.

1 Source question and scope

The source is K. L. Dela Rosa and H. J. Woerdeman, *Location of Ritz values in the numerical range of normal matrices*, arXiv:2004.05288, published in *Linear and Multilinear Algebra* 69 (2021), 2749–2778 [1]. Section 7, page 30 of the arXiv PDF, asks:

Develop similar results for k -Ritz sets, $3 \leq k < n - 1$. Also consider a normal matrix with some eigenvalues in the interior of its numerical range; in that case it is less clear how to characterize $\text{conv } \mathcal{B}_A(\mu_1)$.

The source's Figure 2 shows that its earlier polygonal formula can fail as soon as one eigenvalue is interior. The following is the required source screenshot; it includes both open problems and that numerical example.

This packet solves the complete 4×4 support-envelope problem when the eigenvalues are distinct and noncollinear and the prescribed first Ritz value is not itself an eigenvalue. It also gives a structural formula in arbitrary dimension. It does not solve the higher- k question, the singular case $\mu \in \sigma(A)$, or the search for a comparably simple geometry in dimensions $n \geq 5$.

$$\mathcal{B}_A(\mu_1) = \text{conv}\{w_{123}, w_{12n}, \lambda_3, \dots, \lambda_n\}$$

due to Proposition 5.4. By Proposition 5.6 the assertion follows. $\blacksquare$

7. Concluding remarks

In this study, we considered 2-Ritz sets of a normal matrix having no eigenvalues in the interior of its numerical range. We identified the smallest convex region containing all μ_2 's for which $\{\mu_1, \mu_2\}$ is a 2-Ritz set. An open problem is to develop similar results for k -Ritz sets where $3 \leq k < n - 1$.

Another open problem is to consider a normal matrix with some eigenvalues in the interior of its numerical range. In this case however, it is less clear how to characterize $\text{conv}[\mathcal{B}_A(\mu_1)]$. Figure 2 shows a numerical example where $\text{conv}[\mathcal{B}_A(\mu_1)] \neq \text{conv}\{w_{123}, w_{124}, \lambda_3, \lambda_4\}$.

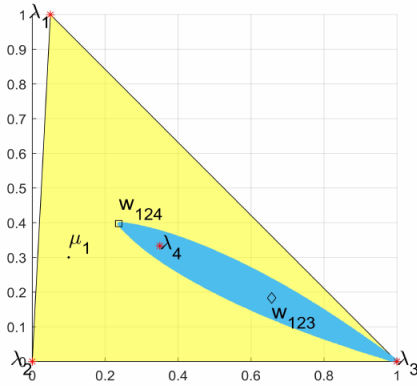


Figure 2.: An example where Theorem 1.1 fails to hold when $\lambda_4 \in \text{conv}\{\lambda_1, \lambda_2, \lambda_3\}$. The blue region is the set $\mathcal{B}_A(\mu_1)$.

Figure 1: The open problems and Figure 2 in [1, p. 30], rendered from the stored source PDF.

2 Setting

For $A \in \mathbb{C}^{n \times n}$ and $\mu \in W(A)$, define

$$\mathcal{B}_A(\mu) = \left\{ \nu \in \mathbb{C} : V^* A V = \begin{bmatrix} \mu & * \\ 0 & \nu \end{bmatrix} \text{ for some } V \in \mathbb{C}^{n \times 2}, V^* V = I_2 \right\}.$$

Unitary similarity lets us take $A = \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Put

$$\mathcal{C}_\Lambda(\mu) = \left\{ t \in \mathbb{R}_+^n : \sum_i t_i = 1, \quad \sum_i t_i \lambda_i = \mu \right\}.$$

For $t \in \mathcal{C}_\Lambda(\mu)$ set $u_i = \sqrt{t_i}$. The condition that $[u, v]$ give an upper triangular compression is precisely

$$v \perp u, \quad v \perp \Lambda u.$$

Consequently, if the columns of F_t form an orthonormal basis of $\{u, \Lambda u\}^\perp$, the standard parametrization in the source is

$$\mathcal{B}_A(\mu) = \bigcup_{t \in \mathcal{C}_\Lambda(\mu)} W(F_t^* \Lambda F_t). \quad (2.1)$$

The choice $u_i \geq 0$ loses no generality: phases can be removed by a diagonal unitary commuting with Λ .

For a direction $\theta \in \mathbb{R}$, write

$$h_i(\theta) = \Re(e^{-i\theta} \lambda_i), \quad H_\theta = \text{diag}(h_1(\theta), \dots, h_n(\theta)).$$

The support of $W(F_t^* \Lambda F_t)$ in that direction is the largest eigenvalue of $F_t^* H_\theta F_t$.

3 Idea of the proof

The apparent obstruction is that (2.1) is a continuum of numerical ranges. The key observation is that every directional Hermitian compression has a single explicit characteristic polynomial. Cauchy–Binet expands the determinant using $(n-2)$ -row minors of F_t . Because F_t is completed by the two-dimensional space spanned by u and Λu , complementary minors turn each coefficient into the elementary weight

$$t_i t_j |\lambda_i - \lambda_j|^2.$$

Thus there is no hidden choice of basis F_t .

For four eigenvalues the three real barycentric constraints leave only one weight parameter s . The support is the maximum of the larger root of a quadratic $P_\theta(s, x)$. Ordinary one-variable calculus now exhausts every maximizer: an endpoint, a differentiable stationary root, or a collision of the two roots. Those three cases are expressed by two endpoint quadratics and the systems $P = P_s = 0$ and $P = P_x = 0$. The proof below makes both reductions exact. The remaining broad problem is not existence of an algorithm, but a simple higher-dimensional geometric description.

4 The support polynomial in all dimensions

Define

$$\delta(t) = \sum_{1 \leq i < j \leq n} t_i t_j |\lambda_i - \lambda_j|^2, \quad (4.1)$$

$$P_\theta(t, x) = \sum_{1 \leq i < j \leq n} t_i t_j |\lambda_i - \lambda_j|^2 \prod_{\ell \notin \{i, j\}} (x - h_\ell(\theta)). \quad (4.2)$$

Theorem 1 (Directional compression polynomial). *Let $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ and let $\mu \in W(\Lambda) \setminus \sigma(\Lambda)$. For every $t \in \mathcal{C}_\Lambda(\mu)$,*

$$\det(xI_{n-2} - F_t^* H_\theta F_t) = \frac{P_\theta(t, x)}{\delta(t)}. \quad (4.3)$$

In particular all zeros of $P_\theta(t, \cdot)$ are real, and

$$h_{\text{conv } \mathcal{B}_\Lambda(\mu)}(\theta) = \max_{t \in \mathcal{C}_\Lambda(\mu)} \max\{x \in \mathbb{R} : P_\theta(t, x) = 0\}. \quad (4.4)$$

Proof. Let $Q = [u, \Lambda u]$. Since $\mu \notin \sigma(\Lambda)$, the two columns of Q are linearly independent for every feasible t : dependence would force all positive-weight eigenvalues to equal μ . Hence

$$E = Q(Q^*Q)^{-1/2}$$

has orthonormal columns and $U = [F_t, E]$ is unitary. Since $xI - F_t^* H_\theta F_t = F_t^*(xI - H_\theta)F_t$, Cauchy–Binet gives

$$\det(xI - F_t^* H_\theta F_t) = \sum_{|I|=n-2} |\det F_t[I, :]|^2 \prod_{\ell \in I} (x - h_\ell(\theta)). \quad (4.5)$$

Complementary minors of the unitary U have equal modulus. If $I^c = \{i, j\}$, then

$$\begin{aligned} |\det F_t[I, :]|^2 &= |\det E[\{i, j\}, :]|^2 \\ &= \frac{|\det Q[\{i, j\}, :]|^2}{\det(Q^*Q)} = \frac{t_i t_j |\lambda_i - \lambda_j|^2}{\det(Q^*Q)}. \end{aligned}$$

Moreover,

$$\begin{aligned} \det(Q^*Q) &= \sum_i t_i |\lambda_i|^2 - \left| \sum_i t_i \lambda_i \right|^2 \\ &= \sum_{i < j} t_i t_j |\lambda_i - \lambda_j|^2 = \delta(t) > 0. \end{aligned} \quad (4.6)$$

Substitution in (4.5) proves (4.3). Real-rootedness follows because the left-hand side is the characteristic polynomial of a Hermitian matrix. Finally, (2.1) and the variational characterization of the largest Hermitian eigenvalue give (4.4). Taking the convex hull does not change a support function. $\square$

5 Exact four-by-four envelope

Assume now that $n = 4$ and the four eigenvalues are distinct and not all collinear. The real 3×4 matrix with rows

$$(1, 1, 1, 1), \quad (\Re \lambda_i)_{i=1}^4, \quad (\Im \lambda_i)_{i=1}^4$$

has rank three. Thus every nonempty $\mathcal{C}_\Lambda(\mu)$ is a possibly degenerate segment

$$t(s) = t^0 + sd, \quad a \leq s \leq b.$$

For fixed θ abbreviate

$$p(s, x) = P_\theta(t(s), x), \quad r_+(s) = \max\{x \in \mathbb{R} : p(s, x) = 0\}.$$

The polynomial p has degree at most two in each variable. Its leading coefficient as a polynomial in x is $\delta(t(s)) > 0$.

Define three sets of real candidate values:

$$\begin{aligned} \mathcal{E}_\theta &= \{x : p(a, x) = 0 \text{ or } p(b, x) = 0\}, \\ \mathcal{S}_\theta &= \{x : \exists s \in [a, b], p(s, x) = p_s(s, x) = 0\}, \\ \mathcal{M}_\theta &= \{x : \exists s \in [a, b], p(s, x) = p_x(s, x) = 0\}. \end{aligned}$$

If $a = b$, interpret $\mathcal{E}_\theta$ as the roots at that single point.

Theorem 2 (Exact 4×4 interior-spectrum envelope). *Let A be a normal 4×4 matrix with four distinct noncollinear eigenvalues, and let $\mu \in W(A) \setminus \sigma(A)$. Then, in every direction θ ,*

$$h_{\text{conv } \mathcal{B}_A(\mu)}(\theta) = \max(\mathcal{E}_\theta \cup \mathcal{S}_\theta \cup \mathcal{M}_\theta). \quad (5.1)$$

Consequently

$$\text{conv } \mathcal{B}_A(\mu) = \bigcap_{\theta \in \mathbb{R}} \left\{ z \in \mathbb{C} : \Re(e^{-i\theta} z) \leq h_{\text{conv } \mathcal{B}_A(\mu)}(\theta) \right\}, \quad (5.2)$$

where each support value is determined by polynomial systems of bidegree at most $(2, 2)$.

Proof. If the segment is a point, (5.1) is immediate. Suppose $a < b$. The two roots of $p(s, \cdot)$ are eigenvalues of the continuous family of 2×2 Hermitian compressions described by Theorem 1; equivalently, the quadratic formula shows directly that r_+ is continuous. It therefore has a maximizer s_* on $[a, b]$.

If s_* is an endpoint, the value $r_+(s_*)$ lies in $\mathcal{E}_\theta$. Suppose $s_* \in (a, b)$. If $r_+(s_*)$ is a simple root, the implicit-function theorem applied to $p(s, r_+(s)) = 0$ gives

$$r'_+(s_*) = -\frac{p_s(s_*, r_+(s_*))}{p_x(s_*, r_+(s_*))} = 0,$$

so the value lies in $\mathcal{S}_\theta$. If it is not simple, it lies in $\mathcal{M}_\theta$. Hence the right side of (5.1) is at least the support in (4.4).

Conversely, every x in any of the three candidate sets is a root of $p(s, \cdot)$ for some feasible s . It is therefore at most $r_+(s)$, which is at most the support. This proves (5.1). Formula (5.2) is the standard supporting-half-plane reconstruction of a compact convex subset of $\mathbb{C}$. $\square$

Corollary 3 (Generic quartic computation). *Write*

$$p(s, x) = A(x)s^2 + B(x)s + C(x) = a(s)x^2 + b(s)x + c(s).$$

Generically, the simple stationary candidates in Theorem 2 are obtained from

$$B(x)^2 - 4A(x)C(x) = 0, \quad s = -\frac{B(x)}{2A(x)} \in (a, b), \quad (5.3)$$

and the multiple-root candidates from

$$b(s)^2 - 4a(s)c(s) = 0, \quad x = -\frac{b(s)}{2a(s)}, \quad s \in (a, b). \quad (5.4)$$

Both displayed polynomial equations have degree at most four. Together with the roots of the endpoint quadratics and feasibility filtering, they give the exact support. In degenerate cases one uses the original systems $p = p_s = 0$ and $p = p_x = 0$; if a stationary component persists, its simple root is constant on that component, and its boundary value is already captured by an endpoint or a multiple-root candidate.

Proof. At a simple stationary solution with $A(x) \neq 0$, the equation $p_s = 2A(x)s + B(x) = 0$ gives the stated s , and substitution into $p = 0$ gives $B^2 - 4AC = 0$. Each of A, B, C has degree at most two in x , so this equation has degree at most four. A multiple root in x is exactly a zero of the quadratic discriminant $b^2 - 4ac$, also of degree at most four in s ; here $a(s) = \delta(t(s)) > 0$. Vanishing leading coefficients are handled without division by retaining the two original polynomial equations. On any positive-dimensional simple-root component of $p = p_s = 0$, implicit differentiation makes the root constant. The final assertion follows by continuity and the classification in Theorem 2. $\square$

6 Verification and failed simplifications

The included script `code/verify_envelope.py` constructs the weight segment, evaluates (4.2), and compares its two roots with a direct orthogonal nullspace compression. With deterministic seed 20260817 it passed 600 such comparisons; the worst absolute root discrepancy was 1.443×10^{-15} . It also numerically optimized directional supports and checked $p_s = 0$ at 45 interior maximizers, with worst normalized residual 5.434×10^{-10} .

For the source-like example

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (i, 0, 1, 0.35 + 0.33i), \quad \mu = 0.10 + 0.30i,$$

the exact segment envelope exceeds the two endpoint ellipses by 0.055372249026 in one tested direction, and 66 of 361 tested directions have an interior maximizing weight. Thus endpoint triangles alone cannot recover the Figure 2 phenomenon. A different tempting relaxation—slicing the unrestricted compression-polynomial coefficient polytope by the fixed root—overestimates support by as much as 0.196837808229 in this example. The code is a consistency audit, not part of the proof.

7 Novelty check, limitations, and review recommendation

A bounded search through 17 August 2026 used the exact source title and authors, forward-citation records, and the phrases “interior eigenvalues”, “ k -Ritz sets”, and “ $\mathcal{B}_A(\mu)$ ”. The closest later work

found is Dela Rosa and Woerdeman’s continuity paper [2], which proves continuity of associated submatrix and Ritz-set maps, not the convex-envelope formula here. Dela Rosa’s 2021 dissertation repeats the open directions. Searches for algebraic Ritz-tuple characterizations found Cauchy-matrix criteria, but no fixed- μ support-polynomial or quartic envelope theorem. No matching solution was found; this is a bounded negative search, not a proof of novelty.

The exact remaining limitations are:

1. Theorem 2 excludes $\mu \in \sigma(A)$, where Q may lose rank and a separate analysis is required.
2. For $n \geq 5$, Theorem 1 gives an exact optimization over a higher-dimensional polytope. Facewise Lagrange-multiplier equations give a finite semialgebraic procedure, but not a concise geometric answer.
3. The source’s separate question about simultaneous k -Ritz sets for $3 \leq k < n - 1$ remains open.

The verdict is **candidate partial, likely valid**. Human review should focus on the minor identity in (4.5)–(4.6) and the exhaustive maximum classification in Theorem 2. The degenerate-component sentence in Corollary 3 is a third review point. The packet deliberately does not promote the result as a full answer to either broad source question.

References

- [1] K. L. Dela Rosa and H. J. Woerdeman, *Location of Ritz values in the numerical range of normal matrices*, Linear and Multilinear Algebra **69** (2021), 2749–2778. doi:10.1080/03081087.2020.1761280; arXiv:2004.05288.
- [2] K. L. Dela Rosa and H. J. Woerdeman, *Continuity of submatrices and Ritz sets associated to a point in the numerical range*, Linear Algebra Appl. **624** (2021), 1–13. doi:10.1016/j.laa.2021.03.031.